PRIYADARSHINI COLLEGE OF ENGINEERING, NAGPUR

Department: Computer Technology   Semester: VIIIth   Section: A and B

CAT-II (2023-24)

Subject	: Social Network		Subject Code	: BTCT801-T
Duration	: 1.5Hrs		Max. Marks	: 35

Note:
1) All questions are compulsory.
2) All questions carry marks as indicated.

			Questions	Marks	CO	BL
Q.1	A	I	Which of the following is the most viral section of the internet? a) Chat Messenger b) Social networking sites c) Tutorial sites d) Chat-rooms	1M	CO3	I
		II	____________ type of sites are known as friend-of-a-friend site. a) Chat Messenger b) Social networking sites c) Tutorial sites d) Chat-rooms	1M	CO3	II
	B		Explain fact equal coin distribution.	5M	CO3	II

OR

Q.2	A	I	What happens when the gold coins distribution game converge? a)all the nodes get equal number of coins b)all the nodes are visited at least once c)each node might have different number of coins d) one node gets all the coins	1M	CO3	II
		II	In a game distributing gold coins across a network, each node begins with an equal number of coins. Following one round of redistribution, the total number of coins across the network remains the same. Which network structure guarantees this outcome? a)Random network b)Scale-free network c)Ring network d) None of the above	1M	CO3	II
	B		Explain Web graph in social network.	5M	CO3	II

			Questions	Marks	CO	BL
Q.3	A	I	If the payoff for working on your Assignment is 0.3 and participating in a Hackathon is 0.5. Five of your friends are participating in the Hackathon and six of your friends are working on the Assignment, what will you opt for? a) Participate in Hackathon b) Work on Assignment	1M	CO4	II

c) Cannot say

	II	Consider the modeling of information cascade in a graph G, where everyone starts with B and there is a small set S of early adopters of A. Given following two statements. S1: Nodes in S keep using A no matter what payoffs tell them to do S2: Nodes outside S keep using B no matter what payoffs tell them to do. Choose the correct option. a)Both S1 and S2 are true b)S1 is true but S2 is false c)S2 is true but S1 is false. d) Both S1 and S2 are false	1M	CO4	II
B		Write a note on : Diffusion in Network	5M	CO4	I
C		Define Power law and explain richer-get richer phenomenon.	7M	CO4	I

OR

Q.4	A	I	What is the density of a cluster? a)fraction of nodes' friends present inside the cluster b)fraction of the node's friends present outside the cluster c)average number of friends inside the cluster d) maximum number of friends inside the cluster	1M	CO4	II
		II	Pick out the reason for Power Law Distribution in real world networks. a) Triadic closure b) Membership closure c) Preferential attachment d) Focal closure	1M	CO4	II
	B		Explain Impact of Commmunities on Diffusion.	5M	CO4	II
	C		Analyze the connection between power law and the spread of epidemics.	7M	CO4	III

Q.5	A	I	Which of the following statements is/are correct? Statement I - SIR model should come to an end after running for a finite number of steps on a network Statement II - SIS model can keep running indefinitely on a network a)I only b)II only c)Both d)None	1M	CO5	III
		II	Assume for a disease 'C', People who are diagnosed in the earlier stage have a high chance of recovery. The recovered people do not have a chance to get infected again. What kind of model does this disease 'C' exhibit? a)SIS b)SIR c) Both SIS and SIR d)None	1M	CO5	II
	B		Explain Epidemics in Social Network in detail.	5M	CO5	III
	C		Compare SIR and SIS spreading models.	7M	CO5	II

OR

Q.6	A I	Select the factors that affect the spread of a contagion. a)density of the contact network b)degree of contagiousness c)both density of network and degree of contagiousness d) none	1M	CO5	II
	II	Suppose the basic reproductive number is estimated to be 0=2.5. A vaccine providing a certain level of immunity is introduced and the new reproductive number is found to be 1=1.8, What is the percentage of immunity provided by the vaccine? a)25% b)28% c)72% d)75%	1M	CO5	II
	B	Illustrate Milgram's Experiment in Detail.	5M	CO5	II
	C	Explain small world phenomenon in social network.	7M	CO5	II

********* Best of Luck **********